

```

GNU nano 0.7.1
#include <stdio.h>
#include <stdlib.h>
#include <string.h>
#include <ctype.h>

#define MAX_TOKENS 50

/* Token structure */
struct Token {
    char value[100];
};

/* Split input into tokens */
int tokenize(char *input, struct Token tokens[]) {
    int count = 0;
    int i = 0;

    while (input[i] != '\0') {
        /* Handle whitespace */
        while (isspace(input[i]))
            i++;

        if (input[i] == '\0')
            break;

        /* Identify delimiters */
        if (input[i] == '|' || input[i] == '<' || input[i] == '>') {
            tokens[count].value[0] = input[i];
            tokens[count].value[1] = '\0';
            count++;
            i++;
        }
        else {
            int j = 0;

            /* Read normal word */
            while (input[i] != '\0' &&
                   !isspace(input[i]) &&
                   input[i] != '|' &&
                   input[i] != '<' &&
                   input[i] != '>') {
                tokens[count].value[j++] = input[i++];
            }

            tokens[count].value[j] = '\0';
            count++;
        }

        if (count >= MAX_TOKENS)
            break;
    }

    return count;
}

/* Validate token stream */
int validateTokens(struct Token tokens[], int count) {
    if (count == 0)
        return 0;

    /* Command cannot start with pipe */
    if (strcmp(tokens[0].value, "|") == 0) {
        printf("Error: Command cannot start with '|' \n");
        return 0;
    }

    /* Command cannot end with pipe */
    if (strcmp(tokens[count - 1].value, "|") == 0) {
        printf("Error: Command cannot end with '|' \n");
        return 0;
    }

    /* Two pipes cannot occur together */
    for (int i = 0; i < count - 1; i++) {
        if (strcmp(tokens[i].value, "|") == 0 &&
            strcmp(tokens[i + 1].value, "|") == 0) {
            printf("Error: Invalid pipe sequence \n");
            return 0;
        }
    }
}

```

Output

```

[ubuntu@ubuntu: ~/OSLab/Skill1]$ gcc skill4.c -o skill4
[ubuntu@ubuntu: ~/OSLab/Skill1]$ ./skill4
Enter command: ls -l /home

Tokens:
Token 1: ls
Token 2: -l
Token 3: /home

Syntax is valid.

Parse / Execution Structure:
COMMAND
  ARGUMENT: ls
  ARGUMENT: -l
  ARGUMENT: /home
[ubuntu@ubuntu: ~/OSLab/Skill1]$

```